Utkarsh Shukla

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Technical Skills

- **Languages:** C++, Java, Python, JavaScript, TypeScript, Golang, SQL
- **Frontend:** React.js, Next.js, HTML5, CSS3, Tailwind CSS
- **Backend & APIs:** Node.js, Express.js, REST APIs, JWT Authentication, WebSockets, Socket.IO
- **Databases:** MongoDB, MySQL, PostgreSQL
- **AI / LLM:** Google Gemini API, RAG, Embeddings
- **Tools & Platforms:** Git, GitHub, GitHub Actions, Docker, Postman, Vercel, Railway
- **Core CS:** Data Structures & Algorithms, OOP, DBMS

Experience

- **TheHXD** *Aug 2026 – Sept 2026*
  Software Development Engineer Intern Remote
  - Developed the frontend for client projects, including **Ceteris Paribus**, using **React.js** and **Next.js**; built responsive pages and reusable UI components from design specs
  - Integrated the UI with backend **REST APIs** (data fetching, forms, error states) and contributed to backend features using **Git** and pull requests

Projects

- **StudyTantra - AI-Powered Learning** | ***Live Link*** *Jan 2026 – Present*
  React.js, Node.js, Express.js, MongoDB, Google Gemini API, RAG, JWT, FSRS [GitHub]
  - Built a full-stack **RAG** pipeline that turns PDFs (up to 10 MB) into study material: per-page text extraction, **Gemini Vision** OCR fallback for scanned files (under 40 characters per page), and sentence-aligned chunks of about 800 tokens with 768-dimension embeddings
  - Implemented grounded chat that retrieves the top 6 chunks by cosine similarity and streams answers over **Server-Sent Events**, with page citations that jump the **react-pdf** viewer to the source page
  - Added **FSRS** spaced repetition targeting 90% retention, weak-topic tracking (topics under 60% accuracy), and an AI exam planner that builds study plans of up to 45 days, alongside summaries, flashcards, and quizzes
  - Secured the API with memory-only 15-minute **JWT** access tokens, hashed rotating refresh tokens in httpOnly cookies with replay detection, **zod** validation, and per-user AI rate limiting (60 requests/hour); deployed on **Vercel** and **Railway**

- **SyncBoard - Real-Time Collaborative Whiteboard** | ***Live Link*** *Jun 2026*
  React.js, Node.js, Socket.IO, WebSockets, HTML5 Canvas [GitHub]
  - Built a multi-user whiteboard with **Socket.IO** and **WebSockets** that syncs 4 drawing tools (pencil, rectangle, line, eraser) on **HTML5 Canvas** to all connected clients in real time
  - Implemented room creation and joining with a live participant list and in-room chat, with a **Node.js** server handling multiple rooms at once

Achievements & Competitive Programming

- **AM Hacks 2.0 Hackathon:** Won **1st Place** (Web3 track) among 400+ teams, hosted at IGDTUW.
- **ICPC Prelims 2026: Institute Rank 1**, **AIR 1013** among 5500+ teams.
- **LeetCode:** Solved **500+ problems**; peak contest rating of **1773**.
- **Codeforces:** Reached **Pupil** (peak rating **1230**); solved **500+ problems**.
- **CodeChef: 2-Star** (peak rating 1500+).
- Qualified for Round 1 of the **Google Big Code Challenge**.

Position of Responsibility

- **Alan Turing Club (DSA & CP Club), Bennett University** *Nov 2025 – Present*
  Technical Head Greater Noida, India
  - Lead technical activities for a club of 50+ members; organized inter-college coding sessions and workshops with **IIIT Sonepat's AlgoZenith Club** on **DSA**, **MERN stack development**, and competitive programming
  - Mentored junior members through 1-on-1 code reviews and structured learning plans; 2 members reached **Pupil** on Codeforces

Education

- **Bennett University** *Sept 2024 – May 2028*
  B.Tech. Computer Science and Engineering | *CGPA: 8.50/10.00* Greater Noida, India